

Getting Started with the Tyrell SDK

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Preface

About This Document

This document provides an introduction to the Tyrell Software Development Kit (SDK) and is intended for engineers that use the Tyrell NEXUS-6 SDK to develop custom application for portable audio devices. In addition, this document describes the procedures required to install the SDK software and hardware. Tyrell recommends that you complete these procedures in the order they are presented in this document.

 **NOTE** This document is intended to be printed using double-sided printing. If you print this document using single-sided printing, some pages will appear blank.

Organization

The following table describes the chapters and appendixes in this document.

Chapter/Appendix	Description
Chapter 1, “Introduction to the Tyrell SDK”	Provides an introduction to the Tyrell NEXUS-6 SDK.
Chapter 2, “Installing the SDK Software and Hardware”	Describes procedures for installing the Tyrell SDK software and hardware.
Chapter 3, “Installing the Green Hills Development Tools”	Describes procedures for installing the Green Hills development tools.
Appendix A, “Common Procedures”	Describes procedures that are common to the Tyrell SDK.
Appendix B, “Development System Default Configurations”	Describes default configurations for the Replicant Engineering Mainboard and the NEXUS-169 CPU Daughtercard.

Document Conventions

Typographic The following table describes the typographic conventions used in this document.

Convention	Description	Example
<i>ABCdef</i>	Identifies book titles, emphasized words or words that appear in the glossary, and command variables.	You <i>must</i> log in as root. C:\> <i>cd</i> <i>directory_name</i> <i>path_name</i>
ABCdef	Identifies graphical user interface items with which you interact.	Click the OK button.
<u>ADCdef</u>	Identifies a hyperlink or URL.	http://www.janaowens.com
ABCdef	Identifies commands, computer-generated output, API elements, and code samples.	Build successful
[]	Indicates optional parameters within a syntax description.	[ALL]
{ }	Indicates a choice between parameters in a syntax description.	{ALL NONE}
	Separates items in a list of choices; used with braces ({ }) in a syntax description.	{ALL NONE}
...	Indicates that you can repeat a series of items one or more times.	<i>path_name</i> ...

Notational The following table describes the notational conventions used in this document.

Type	Symbol	Description
Tip		Provides information that can help you better utilize the product.
NOTE		Provides information that emphasizes the main text.
IMPORTANT		Provides important information required for the successful operation of the product.

Additional Information

The following table identifies additional documentation associated with the Tyrell NEXUS-6 SDK. To download the most recent versions of these documents, visit the Tyrell Extranet.

Document	Description
<i>Tyrell SDK Release Notes</i>	Provides information about the latest Tyrell NEXUS-6 SDK release, including structural and operational changes.
<i>Tyrell SDK Defect Report</i>	Provides a description of all resolved and known defects for the latest Tyrell NEXUS-6 SDK release.
<i>Tyrell SDK Developer's Guide</i>	Provides step-by-step instructions for developing custom applications using the Tyrell NEXUS-6 SDK, as well as an overview of the system architecture and operating system concepts.
<i>Tyrell SDK API Reference</i>	Provides detailed information about the API that is distributed with the Tyrell NEXUS-6 SDK.
<i>Tyrell NEXUS-6 Reference Design Schematics</i>	Provides detailed information about the NEXUS-6.
<i>Tyrell NEXUS-6 Reference Design Schematics</i>	Provides schematics for NEXUS-6-based player designs.

Accessing the Tyrell Extranet

Introduction The Tyrell Extranet provides access to the following product resources:

- ◆ Release notes and defect reports
- ◆ Datasheets and product change notices (PCNs)
- ◆ Product documentation and schematics
- ◆ Source code and upgrades
- ◆ Example applications and application notes
- ◆ Return materials authorization (RMA) and host customization forms

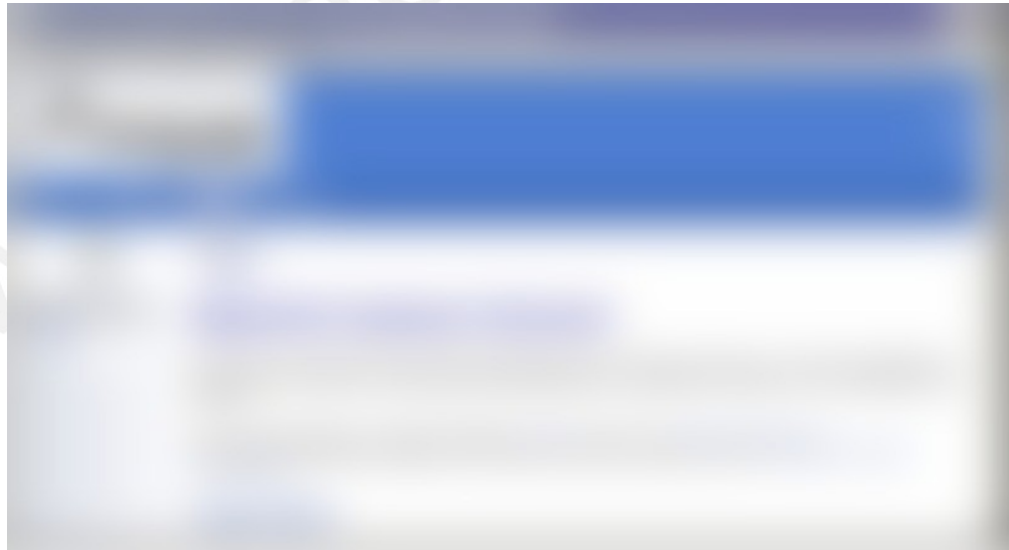
✍ **NOTE** To gain access to the Tyrell Extranet, you must purchase the Tyrell SDK and sign a license agreement. After signing the license agreement, a user name and password will be emailed to you.

Procedure To access the Tyrell Extranet:

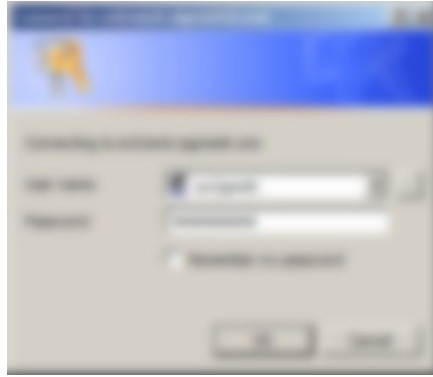
Step 1 Open a browser, and enter the following URL: <http://www.janaowens.com>

Step 2 From the links located on the left side of the page, click **Login**.

The Tyrell Customer Extranet Login page appears.



Step 3 Click **Login Now**.



Step 4 Enter your user name and password, and click **OK**.

 **NOTE** If you are unsure of your user name and password, contact your Tyrell representative.

Step 5 Once the Tyrell Extranet page appears, click **Products**, and then click the **NEXUS-6 Development** link.

Customer Support

Introduction You can request new features, report product defects, or ask general questions by visiting the following URL: <https://www.janaowens.com>

Request New Features To request new features, include the following information in your email:

- ◆ A detailed description of the requested feature, including a description of the user interface
- ◆ Feature specifications
- ◆ Project schedule
- ◆ Datasheet for peripherals

 **NOTE** Tyrell uses this information to prioritize new features.

Report a Defect To report a defect, include the following information in your email:

- ◆ *Tyrell NEXUS-6 SDK version number and part numbers*
Include the version number of your Tyrell NEXUS-6 SDK, and indicate whether any patches have been applied.
- ◆ *CPU product number*
Include the product number listed on the Tyrell CPU chip.
- ◆ *Full description of issue*

Include a detailed description of the issue, as well as answers to the following questions:

- Is the defect included in the *Tyrell SDK Defect Report*?
- Can you reproduce the defect with the latest SDK?
- What steps are required to reproduce the defect?
- ◆ *Debugging information* (if applicable)
Before contacting Tyrell Customer Support, you should attempt to debug the issue. If you are unable to resolve the issue, please include all relevant debugging information in your email submission.
- ◆ *Screenshots* (if applicable)

 **NOTE** When reporting multiple issues, submit only one issue per email.

CHAPTER 1

Introduction to the Tyrell SDK

Introduction This chapter describes the software and hardware that is distributed with the Tyrell SDK. In addition, this chapter provides an overview of the SDK development options.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	Overview	12
B	Software and Development Tools	13
C	NEXUS-6 Development System	14
D	Development Options	15

See Also *Tyrell SDK Developer's Guide*

Section A Overview

Introduction *Proprietary content removed.*

Version Numbers *Proprietary content removed.*

Features *Proprietary content removed.*

For more information about the implementation of these features, refer to the *Tyrell SDK Developer's Guide*.

Deliverables *Proprietary content removed.*

Unit Tests *Proprietary content removed.*

Section B Software and Development Tools

The Tyrell SDK includes the following software and development tools:

Proprietary content removed.

The Tyrell SDK software and development tools are distributed as .zip files and are available for download from the Tyrell Extranet. For information about downloading the Tyrell SDK software and development tools, “Downloading the Tyrell SDK files” on page 18.

Section C NEXUS-6 Development System

The NEXUS-6 Development System includes the following components:

Proprietary content removed.

For more information about the NEXUS-6 Development System, contact your Tyrell representative.

Section D Development Options

The Tyrell SDK provides the following development options:

- ◆ *User Interface Development Tool*

Proprietary content removed.

For information about how to configure and use the User Interface Development Tool, refer to Chapter 2, “Project Management,” in the *Tyrell SDK Developer’s Guide*.

- ◆ *SDK system source code*

Proprietary content removed.

For more information about the system source code, refer to the *Tyrell SDK Developer’s Guide* and the *Tyrell SDK API Reference*.

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CHAPTER 2

Installing the SDK Software and Hardware

Introduction This chapter describes how to install software and hardware that is distributed with the Tyrell NEXUS-6 SDK.

Python Installation *Proprietary content removed.*

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	Downloading the Tyrell SDK files	18
B	Extracting the Tyrell SDK files	19
C	Connecting the Development System	21

See Also *Tyrell SDK Developer's Guide*

Section A Downloading the Tyrell SDK files

Introduction This section describes how to download the Tyrell SDK software and development tools .zip files. The SDK files that you must download depend on the type of application you want to create, as well as your license agreement with Tyrell and other companies; therefore, if you are unsure about which files to download, contact your Tyrell representative. For more information about the Tyrell SDK software and development tools, see “Software and Development Tools” on page 13.

 **NOTE** To complete several procedures in this chapter, an archive utility, such as WinZip, must be installed on your computer. WinZip is used to complete the procedures in this document.

Procedure To download the Tyrell SDK files:

Step 1 Access the Tyrell Extranet as described in “Accessing the Tyrell Extranet” on page 8.

Step 2 From the NEXUS-6 Development page, click the **SDK** link.

Step 3 Download each SDK file by right-clicking the link, and selecting **Save Target As**.

 **IMPORTANT** To download PDDRM or WMDRM libraries, you must have permission to the DRM section of the Tyrell Extranet. If you require access to the DRM section of the Tyrell Extranet, contact your Tyrell representative.

Step 4 Change to the desired directory on your computer, and click **Save**.

Step 5 Repeat the previous steps for each SDK file.

Section B Extracting the Tyrell SDK files

Introduction This section describes how to extract the Tyrell SDK files to the root directory of your local computer.

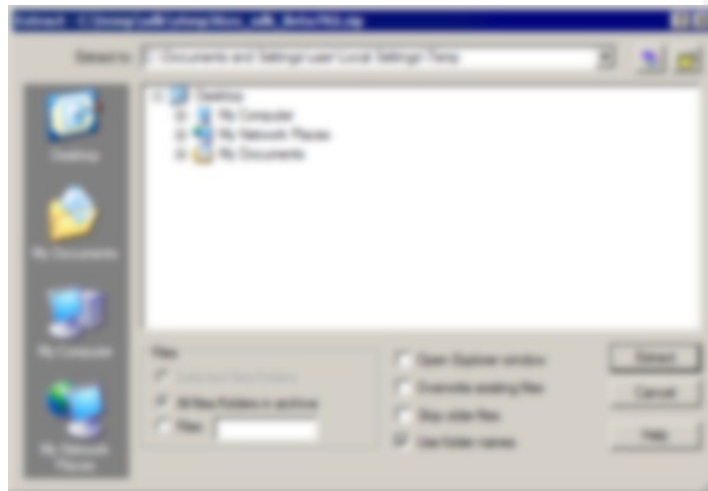
! **IMPORTANT** Patch files may be available for your Tyrell SDK and are intended to replace original SDK files. If you are unsure of whether your SDK requires a patch, contact your Tyrell representative.

Procedure To extract the Tyrell SDK files:

Step 1 Change to the directory in which you saved the files.

Step 2 Right-click a file, select **WinZip**, and then select **Extract to**.

The Extract dialog box appears.



Step 3 In the directory list, select **Local Disk (C)**.

NOTE The example host .zip file contains the example host application installation application and example application binaries. Tyrell recommends that you extract the contents of this file to a temporary directory, such as C:\temp.

Step 4 Select the **Use folder names** option, and click **Extract**.

Step 5 Repeat the previous steps to extract each SDK file.

Result The Tyrell SDK files are extracted to the following location:
C:\NEXUS6_SDK\

! **IMPORTANT** Unless specified otherwise, Tyrell recommends that you extract all SDK files to the location listed above. If the Tyrell SDK directory path exceeds 260 characters or includes spaces, your projects will not compile successfully.

Section C Connecting the Development System

Introduction After you have successfully installed the example host application, you must connect the NEXUS-6 Development System to your computer.

Procedure To connect the NEXUS-6 Development System:

Step 1 Verify the NEXUS-6 Development System is configured according to the default settings. For more information, see Appendix B, “Development System Default Configurations.”

Step 2 *Proprietary content removed.*

Step 3 *Proprietary content removed.*

Step 4 *Proprietary content removed.*

Step 5 *Proprietary content removed.*

Windows automatically recognizes the NEXUS-6 Development System as a Player Recovery Device Class.



NOTE If the Found New Hardware Wizard dialog box appears, select the **Install the software automatically**, and follow the onscreen instructions to complete installation.

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CHAPTER 3

Installing the Green Hills Development Tools

Introduction The Tyrell NEXUS-6 SDK uses the following Green Hills development tools:

- ♦ *Green Hills® MULTI® IDE*—The Green Hills Software MULTI integrated development environment (IDE) is used to create embedded applications and is required to build the Tyrell NEXUS-6 SDK code base, including various components and example projects.
- ♦ *Green Hills Slingshot™ Debug Device*—The MULTI IDE uses the Slingshot debug device to debug and test embedded applications using a USB interface.

This section describes how to install the Green Hills development tools.

! **IMPORTANT** The Tyrell SDK is not backwards compatible to development tools. Refer to the *Tyrell SDK Release Notes* to ensure that you install the correct version of the development tools for your SDK.

Before You Begin Locate the following items:

- ♦ Green Hills Software Installation CD (or installation file)
- ♦ Slingshot Debug Device and installation CD

If you have not received these items, contact your Tyrell representative to request your Tyrell NEXUS-6 SDK.

🔗 NOTE To complete several procedures in this section, an archive utility, such as WinZip, must be installed on your computer.

In this Chapter The following table lists the sections in this chapter.

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Section A Installing the MULTI IDE

Introduction This section describes how to install the MULTI IDE. If a previous version of the MULTI IDE is installed on your computer, you must remove it before proceeding with this procedure.

! IMPORTANT MULTI IDE patches might be available on the Tyrell Extranet. After you complete the installation process, you must apply available patches and reinstall your MULTI license as described in Step 6 of this procedure. If you do not apply the patches, your builds may not complete successfully. For more information, refer to the *Tyrell SDK Release Notes* and the `readme.txt` file located within the patch .zip file.

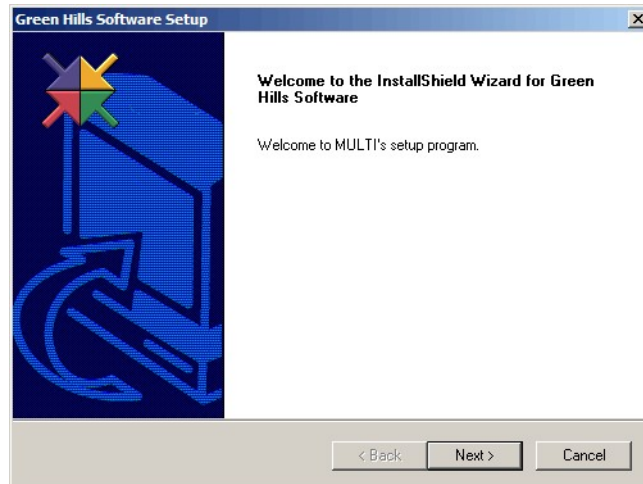
Before You Begin Refer to the *Tyrell SDK Release Notes* to verify that the MULTI IDE is compatible with your version of the Tyrell SDK.

Procedure To install the MULTI IDE:

- Step 1** Download and extract the MULTI IDE installation application and available patches from the Tyrell Extranet:
- A. Access the Tyrell Extranet as described in “Accessing the Tyrell Extranet” on page 8.
 - B. Beneath the NEXUS-6 Development link, click **Green Hills (GHS)**.
 - C. Beneath the MULTI Tools heading, right-click a link, and select **Save Target As**.
 - D. Change to the desired location on your local computer, and click **Save**.
 - E. Repeat the previous step for each file listed under the MULTI Tools heading.
 - F. Change to the directory in which you saved the .zip files, and extract the MULTI Tools file to a location on your local computer.
 - G. Change to the directory in which you extracted the files, locate the installation application (`setup.exe`), and proceed to the next step.

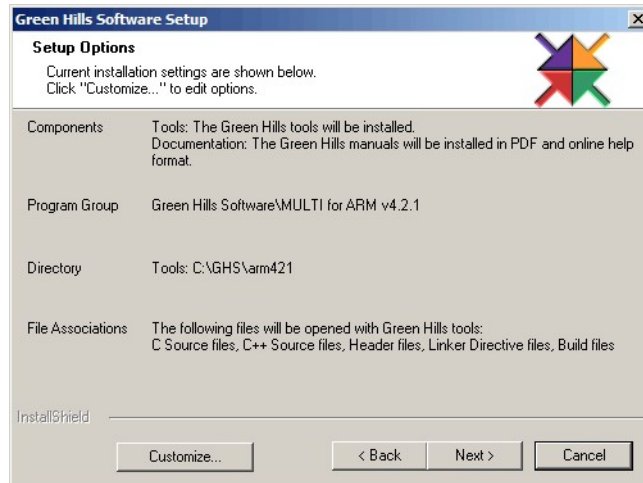
Step 2 Double-click `setup.exe`.

The Green Hills Software Setup dialog box appears.



Step 3 Click **Next**.

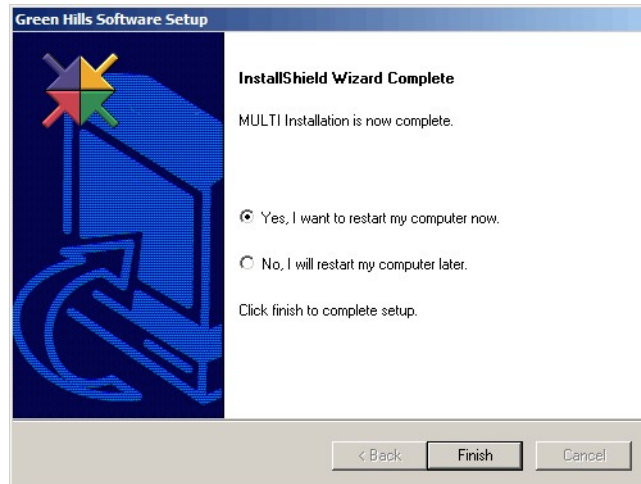
The Setup Options dialog box appears.



Step 4 Click **Next** to install the MULTI IDE software using the default settings. By default, the Green Hills MULTI IDE is installed in the following directory:

C:\GHS

Once installation is complete, you are prompted to restart your computer.



Step 5 Click **Finish** to restart your computer.

Step 6 Install available MULTI IDE software patches.

- A. Locate the .zip file you downloaded earlier in this procedure, and extract the contents of each zip file into the C:\GHS\arm421 directory.
- B. Open a command prompt and change to the C:\GHS\arm421 directory.
- C. Execute the following command for each .iff file:
`gpatch -install patch_xxxx.iff`
- D. Confirm that the patch was installed by opening the patch_information.txt file or by looking in the Add and Remove Programs application, which lists the patch number and the installed directory.
- E. If you already have your MULTI license, reinstall it.

Once you install the MULTI application and patches, you must obtain and install the MULTI license as described in the next sections.

 **NOTE** Green Hills Software version 4.2.3 includes version 4.2.1, plus all applicable patches. If you install version 4.2.3, you do not need to install additional patches.

Section B Obtaining a MULTI license

Introduction This section describes how to obtain a MULTI license. After you submit the required information, Tyrell processes your request and sends you a license file, along with the *Green Hills Software Getting Started Manual*.

 **NOTE** The Green Hills Software MULTI IDE license obtained through Tyrell does not include a simulator or C++ compiler.

Procedure To obtain a MULTI license:

Step 1 Install the MULTI IDE as described in “Installing the MULTI IDE” on page 24.

Step 2 After restarting your computer, open a command-line window. To open a command-line window:

- A. From the Windows Start menu, select **Run**.
- B. In the Run dialog box, type **cmd** in the Open field, and click **OK**.

Step 3 From the command-line, change to the MULTI installation directory. For example, if you used the default settings during installation, change to the following directory:

```
c:\GHS\arm407
```

Step 4 From the GHS\arm407 directory, execute the following command:

```
servecode -g -v
```

Step 5 Copy and paste the output from the command into an email or text file.

Step 6 Send the following information to your Tyrell representative:

- ◆ Servecode output
- ◆ Company name
- ◆ Company address
- ◆ Contact's name
- ◆ Contact's phone number

In response to your email message, Tyrell sends you a license file.

Step 7 Once you receive the license file, install the license as described in “Installing the MULTI license file” on page 29.

Section C Starting the MULTI Launcher

Introduction The MULTI Launcher is a graphical user interface that allows you to install the MULTI license, launch MULTI development tools, access open windows, and manage MULTI workspaces.

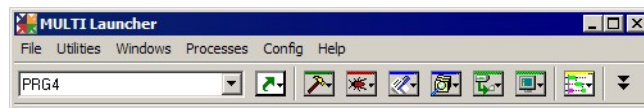
Procedure To start the MULTI Launcher application:

Step 1 From the Windows Start menu, select **Program Files**.

Step 2 Select **Green Hills Software**.

Step 3 Select the **Multi for ARM**, and then select **MULTI**.

The MULTI IDE Launcher appears.



For more information about the MULTI Launcher, refer to the *MULTI: Getting Started* document.

Section D Installing the MULTI license file

Introduction After you receive the license file from Tyrell, you must install the file using the MULTI License Administrator.

Procedure To install the MULTI license file:

Step 1 Open the email you received from Tyrell in response to your license request.

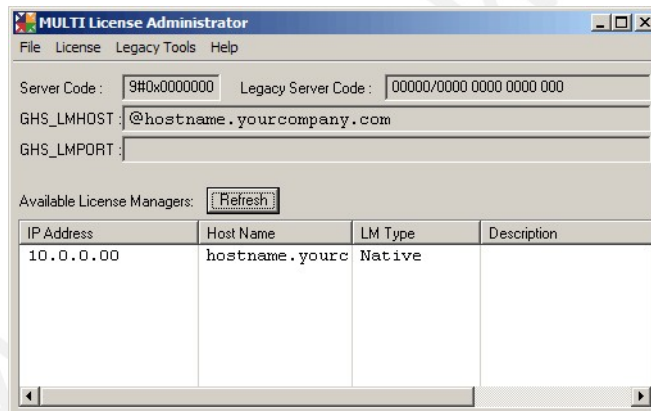
Step 2 Save the attached license file to a location on your local computer.

! **IMPORTANT** Tyrell recommends that you create a backup of your license file by saving an additional copy to a secure location.

Step 3 Start the MULTI Launcher as described in “Starting the MULTI Launcher” on page 28.

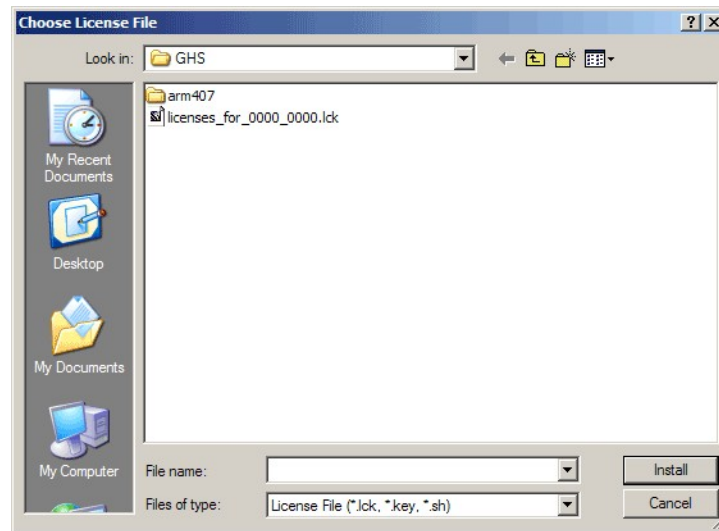
Step 4 From the MULTI Launcher, select **License Administration** from the Utilities menu.

The License Administration dialog box appears.



Step 5 From the MULTI License Administration dialog box, select **Install License File** from the License menu.

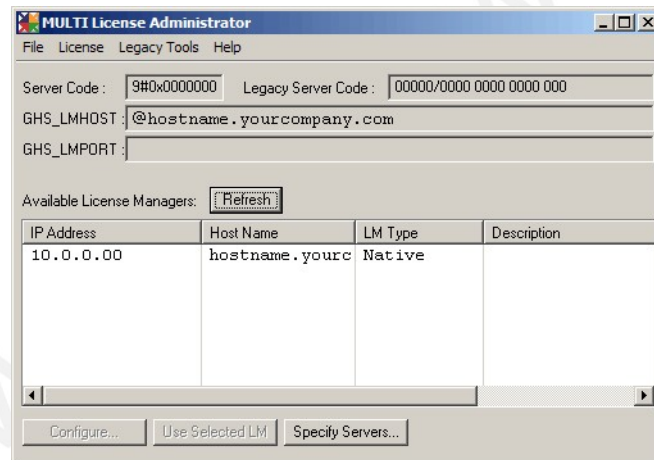
The Choose License File dialog box appears.



Step 6 Navigate to the directory in which you saved the license file.

Step 7 Select the license file, and click **Install**.

Once the license file is installed, license information appears in the MULTI License Administrator dialog box.



Section E Installing the Slingshot debug device

Introduction The Green Hills Slingshot™ is an advanced hardware debug device that enables the MULTI debugger to load, control, debug and test a target system without the need for prior board initialization, an RTOS, or a ROM monitor. This section describes how to install the Slingshot debug device.

For more information about the Slingshot debug device, visit the Green Hills website at the following URL: <https://ghs.com/>

Before You Begin

- ♦ Verify MULTI is installed as described in “Installing the MULTI IDE” on page 24.
- ♦ Verify the NEXUS-6 Development System is configured according to the default configurations described in Appendix B, “Development System Default Configurations.”

Procedure To install the Slingshot debug device:

Step 1 *Proprietary content removed.*

Step 2 Connect the USB type B connector to the Slingshot device.

Step 3 Connect the USB type A connector to an open USB port on your computer.

Step 4 *Proprietary content removed.*

Windows automatically recognizes the new hardware as *Green Hills - Slingshot (tm)*, and the Found New Hardware Wizard dialog box appears.

Step 5 Complete the onscreen instructions to install the new hardware.

Step 6 Open the Device Manager and verify that the *Green Hills - Slingshot (tm)* entry appears under the Universal Serial Bus Controllers heading.

 **NOTE** If Windows does not recognize the hardware, update the driver by opening the Device Manager, right-clicking the device, and selecting **Update Driver**. Slingshot drivers are included in the Green Hills Tools .zip file located on the Tyrell Extranet.

Section F Creating a new probe

Introduction To allow the MULTI IDE to recognize the Slingshot debug device, you must create a new probe using the MULTI Probe Administrator.

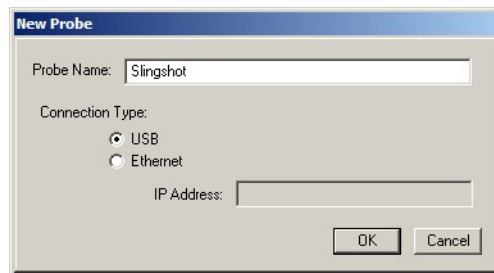
Procedure To create a new probe using the MULTI Probe Administrator:

Step 1 Start the MULTI Launcher as described in “Starting the MULTI Launcher” on page 28.

Step 2 From the Utilities menu, select **Probe Administrator**.

Step 3 In the Probe Administrator dialog box, select **New Probe** from the File menu.

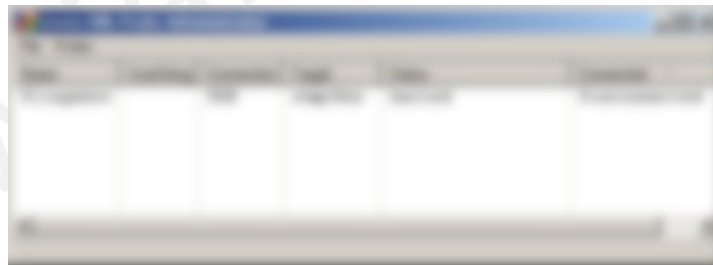
The New Probe dialog box appears.



Step 4 In the Probe Name field, enter a name for the device.

Step 5 Under Connection Type, select the **USB** radio button, and then click **OK**.

The new device appears in the Probe Administrator dialog box.



Step 6 Verify that the Target column displays *nexus6*.



NOTE If the Target column does not display *nexus6*, you must update the probe firmware. For more information, contact your Tyrell representative.

APPENDIX A

Common Procedures

Introduction This appendix describes procedures that are common to the Tyrell SDK.

In this Appendix The following table lists the sections in this appendix.

Section	Topic	Page
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B	Entering recovery mode	35
C	Windows Media DRM 10 Procedures	36
D	Compliance Testing	42

Section A Downloading binaries

- Introduction** The Tyrell SDK example host application is used to download binaries to the development system. This section describes how to download the binaries.
- Before You Begin** Verify the NEXUS-6 Development System is configured according to the default configurations described in Appendix B, “Development System Default Configurations.”
- Procedure** *Proprietary content removed.*

Section B Entering recovery mode

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Issues *Proprietary content removed.*

Resolution Verify your development system is configured as described in Appendix B, “Development System Default Configurations” and then try again to enter recovery mode.

Section C Windows Media DRM 10 Procedures

Introduction This section provides the procedures necessary to obtain from Microsoft Windows Media DRM 10 for Portable Devices and to store the Windows Device Authorization Certificate (DAC) within your custom project.

In this Section The following table lists the topics in this section.

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Obtaining a DACResponse file	38
Creating an unsigned Device Certificate Template	39
Creating a signed Device Certificate Template	40
Creating and storing the build resource files	41

Obtaining Windows Media DRM 10 for Portable Devices

Introduction To distribute WMDRM 10 on devices that can store and play protected content, you must obtain a WMDRM 10 device certificate for Portable Devices.

Procedure To obtain WMDRM 10 for Portable Devices:

Step 1 Request from Microsoft a Windows Media DRM 10 for Devices Development and Interim Product Distribution Agreement by completing the License Request Form located at the following URL: www.microsoft.com

Once Microsoft approves the license agreement, you receive an email from *wmla@microsoft.com* that includes instructions on how to download additional materials from the Microsoft SpecServer, as well as instructions for requesting a device certificate.

! **IMPORTANT** This process may take several weeks to complete. We encourage you to submit your request as soon as possible. You will not be able to use online music services until you obtain a device certificate and complete the procedures described in this section.

Step 2 Access the Microsoft SpecServer using the instructions provide to you by Microsoft.

Step 3 Download the WMDRM 10 OEM installer package.

Step 4 Extract the files to your computer by double-clicking the installer package.

The WMDRM 10 OEM Kit is extracted to the specified location on your computer.

✍ **NOTE** If you used the default settings, the WMDRM 10 OEM Kit is extracted into the WMDRMPD_OEM_Kit directory. The WMDRMPD_OEM_Kit directory name is used within this document to refer to the WMDRM 10 OEM Kit directory.

Obtaining a DACResponse file

Introduction The DACResponse file is an XML file that is used to sign the Device Certificate Template (DCT) described in “Creating an unsigned Device Certificate Template” on page 39. You must obtain the DACResponse file from Microsoft.

 **NOTE** To receive and install a Windows Device Authorization Certificate (DAC), you must execute the Microsoft Windows Media DRM 10 for Devices Final Product Distribution Agreement, which is included with WMDRM 10 for Portable Devices, or be named as a *Third-Party Installer* or as an *Authorized Contractor* under another company’s WMDRM 10 Final Distribution Agreement.

Procedure To obtain a DACResponse file:

Step 1 Complete the Certificate Request Form that is provided with the WMDRM 10 OEM kit.

Step 2 Generate the PrivateKeyFile and the DACRequestFile files.

- A. Click the Windows **Start** button, and select **Run**.
- B. Type **cmd**, and click **OK**.
- C. Change to the WMDRMPD_OEM_Kit\Tools directory, and execute the following command:

```
generatedacrequest.exe -r:DACRequestFile -p:PrivateKeyFile
```

The DACRequestFile and PrivateKeyFile files are created in the WMDRMPD_OEM_Kit\Tools directory. For more information about the GenerateDACRequest tool, refer to the WMDRM_PD.chm file located in the WMDRMPD_OEM_Kit\Documents directory.

Step 3 Email the completed Certificate Request Form and the DACRequestFile to wmla@microsoft.com.

 **IMPORTANT** Do not submit PrivateKeyFile to Microsoft. This file should remain proprietary to your company.

Once the Certificate Request Form and DACRequestFile are processed, Microsoft emails to you a DACResponse file (DACResponse.xml).

Step 4 Place the DACResponse.xml file in the WMDRMPD_OEM_Kit\Tools directory, and proceed to “Creating a signed Device Certificate Template” on page 40.

For more information about this procedure, refer to AcquiringDACResponse.doc located in the WMDRMPD_OEM_Kit\Documents directory.

Creating an unsigned Device Certificate Template

Introduction The Device Certificate Template (DCT) is an XML document that outlines information about your device model and feature set. After the DCT is created, it must be signed by Microsoft using the DACResponse file as described in “Creating a signed Device Certificate Template” on page 40.

Procedure To create an unsigned DCT:

Step 1 Copy the following text from AcquiringDACResponse.doc (located in the WMDRMPD_OEM_Kit\Documents directory) and paste it into a text file.

```
<UNSIGNEDTEMPLATE>
...
</UNSIGNEDTEMPLATE>
```

Step 2 Modify the values of each tag as necessary. Microsoft recommends that you include all information listed; however, you may include only the following tags if desired:

- ◆ <SECURITYLEVEL>
- ◆ <FEATURES>
- ◆ <LIMITS>

! **IMPORTANT** You must change the <SECURITYLEVEL> tag to include the security level provided by Microsoft in the DACResponse file; for example: <SECURITYLEVEL>2000</SECURITYLEVEL>

Step 3 Save the file to the WMDRMPD_OEM_Kit\Tools directory using the following file name: UnsignedTemplate

 **NOTE** This file must be saved using Unicode encoding. To save the file using Unicode encoding, use an application such as Notepad, select **Save As** from the File menu, and then select **Unicode** from the Encoding pull-down menu. In addition, you must remove all tabs and spaces between tags.

For more information about the contents of an unsigned device certificate template and valid values for each tag, refer to the following files located in the WMDRMPD_OEM_Kit\Documents directory:

- ◆ WMDRM_PD.chm
- ◆ AcquiringDACResponse.doc

Creating a signed Device Certificate Template

Introduction Once you create the unsigned DCT, you must sign the DCT using the GenerateDevCertTemplate tool. The GenerateDevCertTemplate tool adds a group certificate to the unsigned template and then signs the template using the PrivateKeyFile. In addition to the signed DCT, DevCertTemplateFile, the GenerateDevCertTemplate tool creates a group private key, OemGroupKey.bin.

! **IMPORTANT** It is critical that you do not disclose private keys to individuals outside your organization.

For more information about the GenerateDevCertTemplate tool, refer to the WMDRM_PD.chm file located in the WMDRMPD_OEM_Kit\Documents directory.

Required Files The following files are used in this procedure:

- ◆ UnsignedTemplate
- ◆ DACResponse.xml
- ◆ PrivateKeyFile

 **NOTE** If you followed the procedures described in this document, these files are located in the WMDRMPD_OEM_Kit\Tools directory.

Procedure To create a signed DCT:

Step 1 Click the Windows **Start** button, and select **Run**.

Step 2 Type **cmd**, and click **OK**.

Step 3 Change to the WMDRMPD_OEM_Kit\Tools directory, and execute the following command:

```
GenerateDevCertTemplate.exe -d:DACResponseFile -u:UnsignedTemplate  
-f:PrivateKeyFile -t:DevCertTemplateFile -k:OemGroupKey.bin
```

The following files are created in the WMDRMPD_OEM_Kit\Tools directory:

- ◆ DevCertTemplateFile
- ◆ OemGroupKey.bin

For more information about these files, refer to AcquiringDACResponse.doc located in the WMDRMPD_OEM_Kit\Documents directory.

Creating and storing the build resource files

Introduction *Proprietary content removed.*

Required Files *Proprietary content removed.*

Procedure *Proprietary content removed.*

Section D Compliance Testing

Introduction *Proprietary content removed.*

**USB
Commands** *Proprietary content removed.*

**PlaysForSure
Requirements** *Proprietary content removed.*

**PlaysForSure
Logo** *Proprietary content removed.*

**PlaysForSure
Testing** *Proprietary content removed.*

**PlaysForSure
Testing with
the SCTL SDK** *Proprietary content removed.*

**PlaysForSure
Test Kit Hints** *Proprietary content removed.*

**Troubleshoot
PlaysForSure** *Proprietary content removed.*

APPENDIX B

Development System Default Configurations

Introduction This appendix describes the default configurations for the Replicant Engineering Mainboard and the different versions of the NEXUS-6 CPU Daughtercard. These default configurations are applied before the NEXUS-6 Development System is shipped.

In this Appendix The following table lists the sections in this appendix.

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A	Replicant Engineering Mainboard	44
B	NEXUS-169 CPU Daughtercard	45
C	NEXUS-100 CPU Daughtercard	46
D	Additional Add-On Daughtercards	47

Section A Replicant Engineering Mainboard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section B NEXUS-169 CPU Daughtercard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section C NEXUS-100 CPU Daughtercard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section D Additional Add-On Daughtercards

Introduction *Proprietary content removed.*

Parallel-to-Serial JTAG Board *Proprietary content removed.*

SDRAM Daughtercard *Proprietary content removed.*

NAND Flash Daughtercard *Proprietary content removed.*

STFM1000 Daughtercard *Proprietary content removed.*

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